

# ACTIVATION-TAIL TURNOVER AS A PREDICTOR OF LAYER-WISE PTQ DEGRADATION: TWO MEASURED CHECKPOINTS, OBSERVED NOT ESTABLISHED

Anonymous authors

Paper under double-blind review

## ABSTRACT

Activation-tail turnover as a predictor of layer-wise PTQ degradation is carrier-dependent in this measured setup: on Qwen3-4B it is a positive relation (point estimate  $\rho \approx 0.59$ ,  $|\rho| \geq 0.3$ , robust to dropping layer 27), while on Qwen3-1.7B it is null ( $\rho \approx -0.11$ ). We measure, on two Qwen3 carriers under 4-bit weight-only HQQ quantization and a calibration-to-deployment prompt-domain shift, the per-block rank correlation between a quantization-free runtime signal — the standardized shift of each block’s activation-tail ratio between the two domains — and the perplexity damage caused by quantizing that one block. The 4B positive relation is itself a depth proxy: turnover is nearly collinear with block depth, and the relation collapses to  $\approx -0.09$  once depth is held fixed, while depth stays predictive given turnover. The decisive readout is bit-identical across three physical hosts (pipeline determinism; effective sample across runs  $n_{\text{eff}} = 1$ ), so we claim determinism, not cross-run independent replication. Because the unit of analysis is a layer of a single model rather than an independent sample, no power analysis backs the primary test, and the pre-registered AUROC gate, where measurable, is not passed, we report a boundary finding about a candidate signal — explicitly exploratory, and not a deployable trigger certificate.

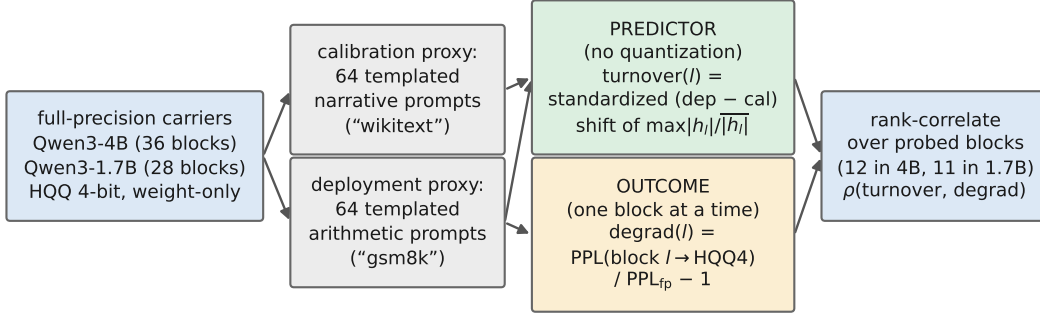
## 1 INTRODUCTION

Serving a post-training-quantized (PTQ) large language model raises a deployment-time question that calibration-time accuracy does not answer: for this traffic, right now, is some block’s quantized forward pass in a regime where it is predictably worse than its full-precision counterpart, so that a fallback to full precision is worth its cost? Reliability benchmarking of PTQ shows that worst-case behaviour degrades under calibration drift and input noise in ways that average accuracy hides (Yuan et al., 2023), and quality metrics have been argued not to be a safety proxy under quantization (Kadadekar, 2026). Existing fallback logic is largely heuristic — dynamic block-level fallback driven by observed numerical overflow during INT8 training (Zhang et al., 2025) — or sidesteps the question by removing calibration from the quantizer altogether (Son et al., 2025). What a runtime trigger would need is a signal that (i) can be computed without running the quantized model and (ii) ranks blocks by how much quantizing them will actually hurt.

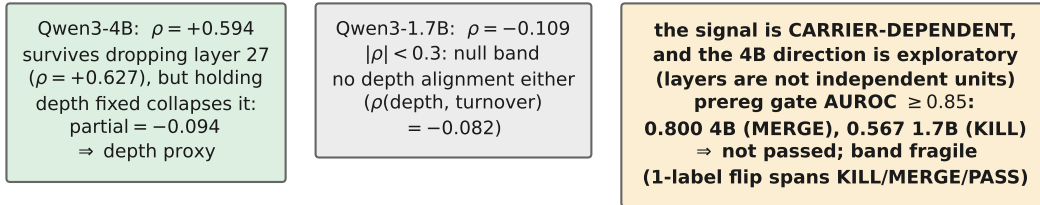
We take one concrete candidate for such a signal and try to falsify it. For each block we measure the *activation-tail turnover*: how much the ratio of the largest to the mean absolute activation entering that block shifts when the input distribution moves from the calibration prompt family to the deployment prompt family, both measured on the full-precision model. As the outcome we take the true per-block damage: quantize exactly that one block to 4-bit HQQ (Badri & Shaji, 2023) and measure the relative increase in deployment perplexity. If the candidate signal is a usable trigger, these two quantities should agree in rank within a model. Figure 1 summarizes the measurement and the answer.

Our measured answer is negative in an informative way, and we report three things. **(1) The relation is carrier-dependent**, not a property of the quantizer or of the domain shift: on Qwen3-4B (Yang et al., 2025) the per-block rank correlation is positive in direction, on Qwen3-1.7B it sits in the null band, and the cross-carrier difference itself is not established by these data (Table 1, Table 2, Fig. 2). **(2) Where the relation exists it is a depth proxy**: turnover on 4B is nearly collinear with block

## How the candidate runtime signal is measured, and what it turns out to be



## Measured readout (single deterministic draw; inference is over layers)



no deployable trigger threshold is claimed — this is a boundary finding about a candidate signal

Figure 1: What is measured and what comes out. The predictor (green) is computed on the full-precision model only, from the domain shift of each block’s activation-tail ratio; the outcome (amber) is measured by quantizing exactly one block at a time and re-measuring deployment-proxy perplexity. The two are then rank-correlated over probed blocks, separately per carrier. The lower row is the measured readout: a positive but depth-proxied relation on Qwen3-4B, a null relation on Qwen3-1.7B, and a pre-registered AUROC gate that is measured and not passed (Limitation F). Reproduce with `python3 experiments/qgate_tailturnover/r31_pipeline_teaer.py`; every value drawn is read at plot time from `R31_PAPER_NUMBERS.json` (sha16 6c26aced50960e6d) and `R40_AUROC_GATE_MASKED.json` (corrected pad-masked labels; Limitation F); figure sha16 c8de713cdda96527. This is a programmatic diagram, not a generated image.

depth, and holding depth fixed collapses the relation while depth remains predictive given turnover (Fig. 3). So the candidate signal, on the carrier where it appears to work, is mostly re-discovering “deeper blocks are more fragile” rather than contributing independent information. **(3) The readout is deterministic but singular:** the decisive payload is bit-identical across three hosts, which certifies the pipeline and lets a reviewer recompute every number, but the effective sample across those runs is one, and the unit of analysis is a layer. We therefore present a boundary finding about signal design and explicitly refuse the deployment-level claim: the pre-registered AUROC gate, where measurable, does not pass (Limitation F), and we propose no trigger threshold.

## 2 WHAT IS MEASURED

### 2.1 PREDICTOR: ACTIVATION-TAIL TURNOVER, COMPUTED WITHOUT QUANTIZATION

Let  $h_l(x)$  be the hidden state entering block  $l$  of the full-precision model on prompt  $x$  (position  $l$  of the hidden-state sequence, with position 0 the embedding output), and define that block’s *tail-mass ratio* on  $x$  as the ratio of the largest to the mean absolute activation over all positions and channels,

$$\text{NTM}_l(x) = \frac{\max |h_l(x)|}{|h_l(x)|}. \quad (1)$$

Given a calibration prompt set  $D_{\text{cal}}$  and a deployment prompt set  $D_{\text{dep}}$ , the turnover of block  $l$  is the standardized shift of this ratio between the two sets,

$$\text{turnover}(l) = \frac{\mathbb{E}_{x \in D_{\text{dep}}} [\text{NTM}_l(x)] - \mathbb{E}_{x \in D_{\text{cal}}} [\text{NTM}_l(x)]}{\frac{1}{2} (\text{sd}_{D_{\text{dep}}} + \text{sd}_{D_{\text{cal}}})}, \quad (2)$$

a Cohen’s- $d$ -style effect size on a per-prompt statistic. Both expectations are taken on the *full-precision* model: the predictor never sees a quantized weight, which is the property a runtime trigger needs, and is also why turnover cannot be a mechanical artifact of the outcome.

## 2.2 OUTCOME: TRUE SINGLE-BLOCK QUANTIZATION DAMAGE

For each probed block  $l$  we rebuild the model from the full-precision checkpoint, replace every linear layer inside block  $l$  — the attention and MLP projections, with a fail-closed assertion that at least six modules were replaced — by a 4-bit HQQ weight-only layer with group size 64, leave every other block and the language-model head in full precision, and measure

$$\text{degrad}(l) = \frac{\text{PPL}_{D_{\text{dep}}}(M_{l \rightarrow \text{HQQ4}})}{\text{PPL}_{D_{\text{dep}}}(M_{\text{fp}})} - 1. \quad (3)$$

The quantity of interest is then the rank correlation  $\rho(\text{turnover}, \text{degrad})$  over probed blocks of one model: does a block whose activation tail turns over more also lose more when it, specifically, is quantized?

## 2.3 CARRIERS, DOMAIN PROXIES, PROBE RULE

Two carriers of the Qwen3 family (Yang et al., 2025) are measured, Qwen3-4B (36 blocks) and Qwen3-1.7B (28 blocks), in `bfloat16`, on one NVIDIA H20-3e device per cell. The probe rule is the fixed index list `[0, 1, 4, 7, 12, 17, 22, 27, -4, -3, -2, -1]` resolved against the block count and de-duplicated, giving 12 probed blocks on 4B and 11 on 1.7B, i.e. early, middle and the four deepest blocks; the largest probed index therefore pins the block count of each carrier. The two prompt families are 64 programmatically templated sentences each, one narrative and one arithmetic; they are labelled `wikitext` and `gsm8k` in the run configuration but are *not* those corpora, and Limitation C11 states exactly what they are. Tail statistics use up to 192 tokens per prompt and perplexity up to 256 tokens, with identical batching for the full-precision and the single-block-quantized models. Each cell writes one row containing the probed layer set, per-block turnover, full-precision deployment perplexity and per-block degradation; the decisive cells are `4B_hqq4_gsm8k` and `1.7B_hqq4_gsm8k` of one registered run configuration, replayed under a second trial on a different physical host (run, trial and host identifiers are withheld for double-blind review and given in the anonymous manifest).

## 2.4 ESTIMATORS

Within a carrier we use the Spearman rank correlation over probed blocks. Uncertainty is reported two ways on purpose — a Fisher- $z$  interval with  $\text{SE} = 1/\sqrt{n-3}$  and a percentile bootstrap resampling blocks — because at these  $n$  the two families disagree about whether zero is excluded on 4B (Table 2); the same holds for a two-sided permutation  $p$ , which lands on either side of 0.05 depending on convention. We therefore condition no claim on a significance threshold, and instead report the point estimate together with leave-one-block-out (LOO) resampling and the pre-registered drop-block-27 re-check. Partial rank correlations are computed by residualizing ranks on ranks; they agree with the closed-form first-order partial correlation to better than  $5 \times 10^{-4}$  on both carriers, so the depth adjudication in §3.2 does not depend on that implementation choice. Cross-carrier difference is summarized as  $\Delta z$  under Fisher’s transform with a bootstrap interval. An independent re-implementation of the estimators by a separate process (NumPy rank/tie handling, its own bootstrap and a fixed-seed permutation null with  $B=20000$ ) reproduces  $\rho$  on both carriers to the digits printed here: the readout is estimator-independent, which is a different and weaker property than being data-independent.

# 3 RESULTS

## 3.1 THE RELATION IS CARRIER-DEPENDENT

Table 1 gives the readout for both carriers and Fig. 2 the underlying per-block scatter; the full Pearson/rank, CI, LOO and robustness series for the same cells is in Table 2, Panel A. On Qwen3-4B the rank (Spearman) correlation between turnover and single-block degradation is positive

Table 1: Carrier-dependent heterogeneity of activation-tail turnover as a predictor of per-layer PTQ degradation (HQQ4 weight-only, arithmetic deployment proxy). Writing  $\tau$  for turnover (Eq. 2),  $\delta$  for single-block degradation (Eq. 3) and  $d$  for the block index, all  $\rho$  are rank (Spearman) correlations and  $\rho(\cdot, \cdot | \cdot)$  is a partial rank correlation, taken over *probed blocks of the same model* — the unit of analysis is a layer, not an independent sample (C4, C9). On 4B the raw relation is strong but  $\tau$  is nearly collinear with  $d$  and the relation collapses once  $d$  is held fixed, i.e. it is a depth proxy; on 1.7B there is neither a raw relation nor a depth alignment to proxy for.

| Carrier    | probed blocks | $\rho(\tau, \delta)$ | $\rho(d, \delta)$ | $\rho(\tau, d)$ | $\rho(\tau, \delta   d)$ | $\rho(d, \delta   \tau)$ | PPL <sub>fp</sub> on dep. proxy |
|------------|---------------|----------------------|-------------------|-----------------|--------------------------|--------------------------|---------------------------------|
| Qwen3-4B   | 12 of 36      | +0.594               | +0.685            | +0.909          | −0.094                   | +0.433                   | 18.953                          |
| Qwen3-1.7B | 11 of 28      | −0.109               | +0.109            | −0.082          | −0.101                   | +0.101                   | 19.597                          |

*Reconstruction.* HQQ4 weight-only; calibration proxy = templated narrative prompts, deployment proxy = templated arithmetic prompts,  $n_{\text{cal}} = n_{\text{dep}} = 64$  (C11 states what these proxies are and are not). PPL<sub>fp</sub> is the full-precision deployment-proxy perplexity, the denominator of every degradation value. Probed-block lists, analysis JSON sha16s, and the double-blind-suppressed provenance (device / trial / host) are consolidated in Appendix Table 3. Uncertainty, robustness and power for these same cells are in Table 2.

( $\rho = +0.594$ ; Table 1, Table 2, Panel A), with its Pearson analogue clearing the pre-registered PASS band; on Qwen3-1.7B the relation is null ( $\rho = -0.109$ ) and its Pearson analogue falls in the pre-registered KILL band. The pre-registered rule and both verdicts are in Appendix Table 4; the rank/CI/LOO/robustness series for both carriers is in Table 2, Panel A. Because the pre-registration defined its PASS/KILL rule over *eight* cells of which only two were run, we report these two cells as a boundary observation, not as a pass of that rule (Limitation C12). The cross-carrier difference points in the sign-flip direction ( $\Delta z = +0.79$ ) but, with two cells and a seed-sensitive interval whose lower edge grazes zero, is observed not established (Table 2, Panel B).

Two honest features of the outcome variable bound how much any of this can mean. First, the damage is extremely concentrated: in each carrier a single block dominates the readout and the largest remaining effect is several times smaller, while roughly half the probed blocks land slightly *below* the full-precision baseline (the concentration rows of Table 2, Panel A). A rank statistic over 11–12 such points is fragile by construction, which is why we report LOO rather than a single interval; on 4B the LOO range stays positive — the relation is not the artifact of one block, but it is carried by a handful of them. Second, the 1.7B null is not a measurement failure: its dominant block is just as damaged, it simply is not the block whose tail turns over most.

### 3.2 WHERE THE RELATION EXISTS, IT IS A DEPTH PROXY

The 4B relation does not survive conditioning on depth. Turnover and block index are nearly collinear, depth by itself ranks damage about as well as turnover does, and the partial rank correlation of turnover with damage given depth collapses toward zero while the partial correlation of depth with damage given turnover stays positive (Table 1; the partials’ layer-resampling bootstrap 95% CIs and permutation  $p$ , which are wide and both span zero, are in Appendix Table 3; cf. R34\_PARTIAL\_CI\_BOOTSTRAP.json; Fig. 3). At  $n=12$  layer-level points this partial contrast is *directionally* consistent but not statistically resolved, while the leave-one-block-out ranges are tighter and point the same way. Re-weighting the probed sample to uniform depth coverage — removing the probe rule’s deliberate deep-decile dominance — lowers but does not erase the depth correlations: the headline falls to  $\rho(\tau, \delta) = 0.401$  once the deep decile ceases to dominate, and the depth correlations shrink to the same degree (Table 2, Panel A, re-weighted rows; R37\_DEPTH\_REWEIGHT\_GATE3). The depth proxy therefore survives the probe-bias correction, though part of its magnitude is probe-induced (Limitation E). Read together with the LOO range, the structure is not a single outlier but a depth-graded pile-up: on this carrier the candidate signal carries approximately the information already available from the block index, and we therefore do not claim that turnover independently predicts quantization damage. On 1.7B there is nothing to mediate — depth and damage are essentially unrelated (Table 1), and depth and turnover are unrelated as well — which is consistent with the raw null and shows that the depth structure itself, not just the signal, is carrier-dependent. A mechanistic reading (weight-only quantization damage concentrating where the residual stream has saturated) is directional only: with 11–12 layer-level points we can locate the confound, not the cause.

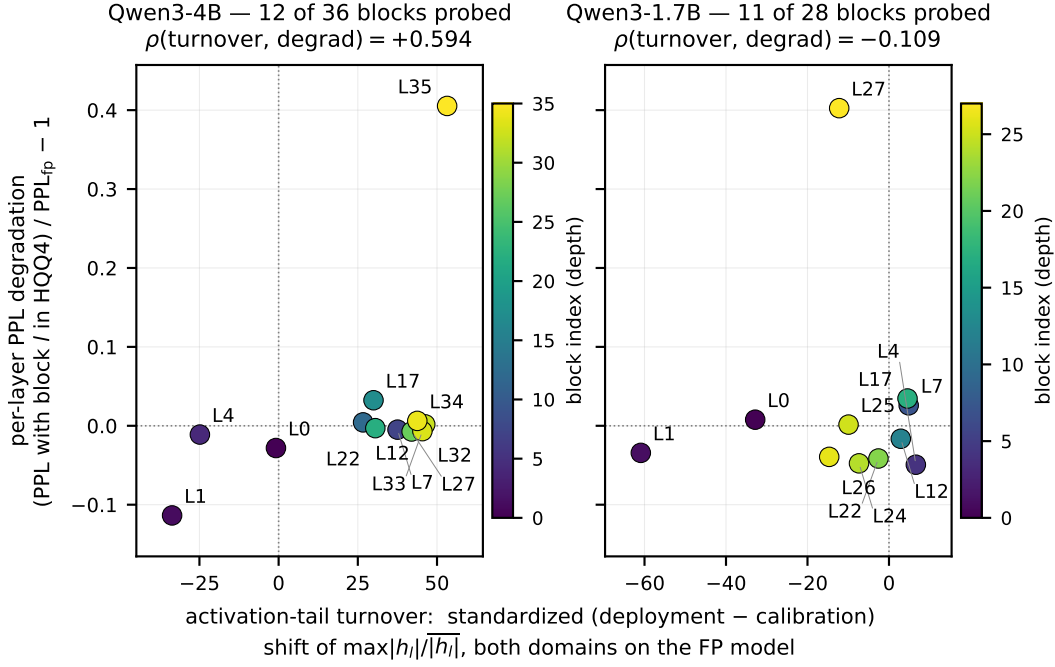


Figure 2: Per-block activation-tail turnover (Eq. 2) against single-block PTQ degradation (Eq. 3), one point per probed block, coloured by block index. Left: Qwen3-4B, where turnover and damage both concentrate in the deepest blocks (yellow, upper right) — this alignment is the depth proxy quantified in §3.2. Right: Qwen3-1.7B, where the deepest probed blocks are spread across the turnover axis and the most damaged block (27) is not the one with the largest turnover. Axis labels state the estimator exactly; note that turnover is a deployment-minus-calibration shift measured on the full-precision model, in pooled standard-deviation units. Reproduce with `python3 experiments/qgate_tailturnover/r31_figure1_scatter_v2.py` (0 GPU) from the cells of Table 1; figure sha16 7a87da7bde09c2b.

### 3.3 DETERMINISM OF THE READOUT, AND WHAT IT IS NOT

The decisive cells were executed three times on three different physical hosts and two trials. The probed layer set, every turnover value, every per-block degradation value and the full-precision deployment perplexity agree exactly across all three runs ( $\max|\Delta| = 0$  in each field; digests in Table 2). This is worth stating because it makes the whole readout replayable by a reviewer at the same point, and because it rules out host- or scheduler-dependent numerical drift. It is equally important what it does not establish: the three runs are a deterministic replay of a single draw, not independent samples, so the effective sample across runs is  $n_{\text{eff}} = 1$  and every interval in this paper is a statement about resampling *layers*, not runs. We consequently never describe this as cross-run independent replication.

## 4 LIMITATIONS

The following are binding on how the results above may be read; several of them were fixed as disclosure requirements before this manuscript was written, and they are the reason the contribution is stated as a boundary finding.

- **(C4) The unit of analysis is a layer, not an independent sample.** Each carrier contributes 12 or 11 probed blocks of one model, and the two carriers share the Qwen3 architecture. Layers within a model are neither independent nor exchangeable, so every interval, partial correlation and power figure here is a layer-level statement carrying a pseudo-replication risk, and none of them supports a population-level claim about models.
- **(C5) The confidence-interval family changes the verdict’s apparent significance.** On 4B the Fisher- $z$  interval excludes zero while the percentile bootstrap contains it, and the two-sided  $p$  moves across 0.05 with the convention used (Table 2). We report the point estimate,

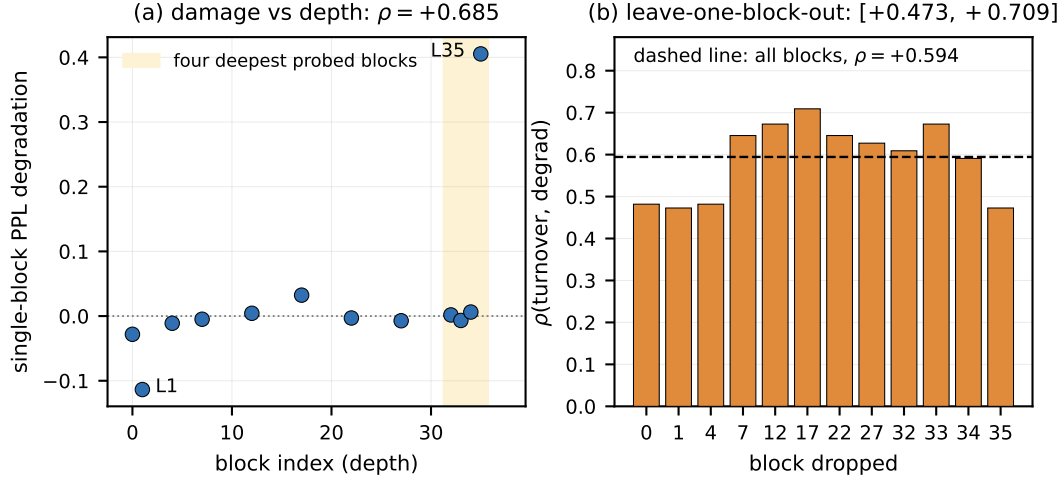


Figure 3: Depth attribution on Qwen3-4B. (a) Single-block damage against block index, with the four deepest probed blocks shaded: the rank relation to depth is  $+0.685$ , and the deep zone contains the one block whose quantization is genuinely expensive (block 35) — but also three blocks that are essentially free, which is why depth is a coarse predictor rather than a law. (b) Leave-one-block-out re-computation of  $\rho(\text{turnover, degrad})$ : every bar stays positive and the minimum is reached exactly when the dominant block 35 (or block 1, the largest negative) is dropped, so no single block manufactures the relation, and none of it survives conditioning on depth either (Table 1). Reproduce with `python3 experiments/qgate_tailturnover/r31_depth_attribution_v2.py` (0 GPU; asserts the drawn LOO range against the registered one); figure sha16 e152c9265ebd45bc, backing analysis JSON sha16 fd1bea221759eb4c.

the drop-block-27 re-check and the LOO range instead, and no classification in this paper is conditioned on zero-exclusion or on  $p < 0.05$ .

- **(C6) A second analyzer in the repository is configuration-mismatched.** One analysis script hard-codes a different results directory and a different cell pair than the cells actually executed. The readout here is computed directly with the same single-source estimator functions as the primary analyzer, so nothing in this paper depends on the mismatched script; we disclose it so that “the analyzer validated it” cannot be inferred.
- **(C7) The pre-registered safety gate’s second half ( $\geq 30\%$  damage gain) was not measured.** The pre-registration’s PASS condition was an AUROC  $\geq 0.85$  trigger *together with* a  $\geq 30\%$  reduction in deployment damage over a static clip baseline; the AUROC half is computed in Limitation F (and does not pass), while the damage-reduction half requires a fallback-baseline simulation and was not run. This paper therefore makes no claim of having passed the registered gate.
- **(F) The pre-registered AUROC gate is measurable here and is not passed, and its band classification is fragile.** The label rule, band definitions, and the AUROC half computed on the corrected pad-masked labels are in Appendix Table 4 (0 GPU; R40\_AUROC\_GATE\_MASKED); the 4B AUROC half is 0.800 (MERGE, not PASS) and the 1.7B half is 0.567 (KILL). This corrects the earlier readout (Limitation C11): under the pad-polluted labels the 4B AUROC appeared lower (0.743) and the trivial depth baseline (block index) *appeared* to beat it; on the masked labels, which the depth-proxy reading in §3.2 is built on, the 4B depth baseline is 0.686 vs. turnover 0.800, so the AUROC endpoint provides *no* additional baseline-beating basis — the depth-proxy reading rests on the partial correlations there, not on this endpoint. Two caveats bind the gate: the damage-gain half ( $\geq 30\%$ ) needs a fallback-baseline simulation and was not measured (C7), and the band verdict is fragile — a single label flip moves the 4B AUROC across the KILL/MERGE/PASS span, and the 4B positive count rests on one near-zero layer (C12). Net: the gate is not passed.
- **(C8) The cross-carrier sign difference is not established, and the second-domain extension is a decision record, not a result.** The  $\Delta z$  bootstrap interval is a two-cell statement whose lower edge grazes zero and is unstable under resampling ( $\Delta z = +0.79$ ; Table 2, Panel B and the seed-envelope note below it — it does not robustly exclude zero —

Table 2: Uncertainty, robustness and design-power ledger. Panel A is per carrier; Panel B is cross-carrier. Every interval is over *probed blocks of one model*, so it quantifies sampling of layers, not of independent runs or models (C4). The two CI families disagree about zero-exclusion on 4B; the verdict deliberately does not rest on either (C5). The *projected* column of Panel B is analytic design planning for two unexecuted zh cells (C8), not a measured result.

| <b>Panel A — per-carrier readout</b>                                     |                    |                   |
|--------------------------------------------------------------------------|--------------------|-------------------|
| Quantity (over probed blocks)                                            | Qwen3-4B           | Qwen3-1.7B        |
| $\rho$ (turnover, degradation), rank                                     | +0.594             | −0.109            |
| Pearson $r$ (turnover, degradation)                                      | +0.490             | +0.021            |
| Fisher- $z$ 95% CI on $\rho$                                             | [+0.031, +0.871]   | [−0.665, +0.525]  |
| Bootstrap percentile 95% CI on $\rho$                                    | [−0.085, +0.921]   | [−0.718, +0.617]  |
| Permutation $p$ (two-sided, $B=20000$ )                                  | 0.045              | 0.749             |
| $\rho$ after dropping block 27                                           | +0.627             | −0.006            |
| Leave-one-block-out range of $\rho$                                      | [+0.473, +0.709]   | [−0.273, +0.188]  |
| Largest single-block degradation                                         | +0.405 (block 35)  | +0.403 (block 27) |
| Second-largest  degradation                                              | 0.114 (block 1)    | 0.049 (block 4)   |
| Probed blocks below the FP baseline                                      | 7 of 12            | 6 of 11           |
| $\rho(\tau, \delta)$ , depth-reweighted (4B)                             | 0.401              | —                 |
| $\rho(d, \delta)$ , depth-reweighted (4B)                                | 0.539              | —                 |
| $\rho(\tau, d)$ , depth-reweighted (4B)                                  | 0.894              | —                 |
| <b>Panel B — cross-carrier difference and design power (layer-level)</b> |                    |                   |
| Quantity                                                                 | measured (2 cells) | projected (+zh)   |
| $\Delta z = z(4B) - z(1.7B)$                                             | 0.794              | —                 |
| $\Delta z$ boot. 95% CI, lower edge over 8 seeds                         | [−0.04, +0.70]     | —                 |
| $\Delta z$ boot. 95% CI, upper edge over 8 seeds                         | [+0.21, +1.39]     | —                 |
| Seeds whose $\Delta z$ CI excludes 0                                     | 7 of 8             | —                 |
| Fisher SE of $\Delta z$                                                  | 0.486              | 0.317             |
| Power to detect $ \Delta z =0.794$                                       | 0.37               | 0.71              |
| Power to detect the 4B depth effect ( $\rho=0.685$ )                     | 0.71               | 0.97              |
| Smallest $ \Delta z $ detectable at 80% power                            | 1.36               | 0.89              |
| Registered gate AUROC (masked) <sup>(F)</sup>                            | 0.800              | 0.567             |
| Noise floor (max within-cell std)                                        | 0.013              | 0.019             |
| $ \delta  \leq$ one noise width                                          | 7/12               | 4/11              |
| Masked rank-repro <sup>(C11)</sup>                                       | 0.74               | 0.88              |
| Runs / hosts of the decisive readout, max $ \Delta $                     | 3 / 3, 0           | —                 |

*Reconstruction and method sensitivity.* All values: python3 experiments/qgate\_tailturnover/r3l\_paper\_numbers.py → R3L\_PAPER\_NUMBERS.json (sha16 6c26aced50960e6d). The bootstrap draw-/convention-sensitivity notes, the  $\Delta z$  seed envelope, the depth-reweight note, the pre-registered correlation rule, and the partial- $\rho$  bootstrap CIs are consolidated in Appendix Table 3; the two pre-registered gate bands and their verdicts are in Appendix Table 4. Effective cross-run sample is 1 (C9); the two CI families disagree about zero on 4B (C5); a single label flip moves the 4B AUROC across [0.639, 0.889], spanning KILL/MERGE/PASS (F, C12).

R35\_DELTA\_Z\_CI\_FIX.json). It is therefore observed, not established. Adding the two unexecuted zh cells would tighten the Fisher SE and raise the power to detect the observed  $|\Delta z|$ , but even so would remain short of 0.80 power for the sign flip, which would need a larger  $|\Delta z|$  (Panel B and its footnote hold the exact numbers); their real value would be testing whether the 4B depth structure reappears in a second domain (projected power > 0.8, Panel B), not pinning the sign flip. Those cells were not run (estimated 0.5–1.5 GPU-h, pending budget approval); the projected column of Table 2 is analytic design planning only.

- **(C9) Bit-identical replay is determinism, not independent replication.** The agreement across three hosts and trials demonstrates pipeline determinism with an effective cross-run sample of 1. We write determinism or bit-exactness, never cross-run independent replication.
- **(C10) No power analysis backs the primary test; the 4B direction is exploratory.** No power analysis was performed for  $\rho$ (turnover, degrad) itself; the only power numbers available (Table 2, Panel B) are layer-level design projections for the cross-carrier difference and for the depth effect. The 4B positive direction is accordingly reported as exploratory, not confirmatory: should any layer-level power analysis find the primary test underpowered, that result must be read as an exploratory observation and the confirmatory-sounding vocabulary

dropped. Because the samples are layers, we make no trigger-threshold or deployment-level claim from it under any circumstance.

- **(C11) The domain shift is a templated prompt-family shift; the headline survives pad-masked perplexity.** The families named `wikitext/gsm8k` are not the WikiText-2/GSM8K corpora but 64 single-template sentences each (a record-filing and an arithmetic word problem), so the shift is a clean low-diversity proxy with reproduction untested on real traffic. The primary analysis did not mask EOS-padding out of the loss; because that inflates absolute PPL, we re-derived every per-block degradation excluding pad positions for all 12/11 probed layers (`masked_from_log`): pad fraction 0.004; the damage ranking is preserved and the 4B headline  $\rho(\text{turnover}, \text{damage})$  falls to 0.559 after masking (1.7B stays null / sign-unstable; masked rank-repro rows in Table 2, Panel A). Absolute perplexities in Table 1 are not compared with published values.
- **(C12) Effects are of the same order as the outcome’s reproduction noise except at one or two dominant blocks, and only 2 of the pre-registered 8 cells were run.** The measured within-cell reproduction spreads of near-zero blocks are milliscale on both carriers (five-arm re-measures, `R39_NOISE_RANK_SYNTHESIS`; noise-floor rows in Table 2, Panel A), yet over half the 4B probed blocks have  $|\delta|$  within one maximum noise width of zero, and although the masked re-compute keeps the ranking, the 4B headline still rests on one or two dominant blocks (drops to 0.43 with the maximum-damage block removed). Separately, the pre-registration defined PASS/KILL over eight cells (2 carriers  $\times$  2 quantizers  $\times$  2 shifts,  $\leq 6$  GPU·h); this study ran only two (one HQQ/4-bit quantizer, one shift, Scope), so the registered eight-cell rule is not directly evaluable — we report a two-cell boundary observation and claim neither a pass nor a kill of the registered rule.
- **(E) The deep-oversampling probe rule makes the depth structure partly a design artifact.** The fixed probe rule  $[0, 1, 4, 7, 12, 17, 22, 27, -4, -3, -2, -1]$  deliberately places four of the twelve probed blocks in the deepest decile, so the depth-graded pile-up of §3.2 is in part a consequence of where we chose to probe rather than of the underlying per-layer damage distribution alone. We therefore read the depth result as a *descriptive* statement about this probe design, not as evidence about the uninspected layers, and the inference we draw is limited to the probed blocks. We quantify this probe bias on the measured sample by re-weighting each probed block to uniform depth coverage (`R37_DEPTH_REWEIGHT_GATE3`): on 4B the headline becomes  $\rho(\tau, \delta) = 0.401$  when the deep decile ceases to dominate, and the depth correlations shrink to the same degree (Table 2, Panel A, re-weighted rows) — so the depth proxy survives re-weighting but is partly inflated by the probe rule. Re-running the single-block-quantized outcome for the unprobed blocks (a depth-uniform or full-layer probe) remains the more expensive step beyond this study’s budget and is not performed here.
- **Scope.** Two carriers of one family, one quantizer, one bit width, one prompt-shift pair, one outcome metric. The relation we do find is a depth proxy, so no independent-mechanism claim is made; and since the second carrier is null, we cannot even say the signal is generally weak-but-real rather than absent.

## 5 RELATED WORK

Runtime-safe quantization triggering has been approached from the numerical side, with block-level fallback to higher precision when overflow is detected during INT8 training (Zhang et al., 2025), and from the quantizer side, by removing the calibration dependency that creates drift sensitivity in the first place (Son et al., 2025); neither certifies a deployment-time predictor of per-layer damage, which is what we test. Reliability studies of PTQ motivate exactly that test by showing how much worst-case behaviour moves under calibration and input shift (Yuan et al., 2023), and by arguing that task-quality metrics do not stand in for safety once weights are quantized (Kadadekar, 2026). Two further bodies of work bound what our result can mean. Layer-sensitivity and mixed-precision schemes already treat blocks as individually vulnerable and allocate width by per-layer sensitivity (Frantar et al., 2023; Lin et al., 2024; Dettmers et al., 2022; Li et al., 2025), but do so from an *offline* sensitivity table rather than from a quantization-free *runtime* signal; and the activation-outlier story that motivated activation-aware quantization has been shown to have a diminishing, calibration-set-dependent effect in modern LLMs (Paglieri et al., 2024). Our per-block framing asks



the deployment-time question both of them leave open — can the *shift* of a tail statistic rank per-block damage? — and finds that where it appears to, the ranking is inherited from block depth, so any candidate runtime signal must first beat the trivial structural baseline of block index.

## 6 CONCLUSION

Activation-tail turnover is a carrier-dependent predictor of per-block PTQ damage: positive in direction on Qwen3-4B, null on Qwen3-1.7B, and, where positive, a proxy for block depth rather than an independent signal. The readout is deterministic to the bit across three hosts, which makes it auditable, and singular, which makes it exploratory. Two practical implications follow for trigger design, and both are about method rather than about this particular statistic. A single hand-picked activation statistic should not be promoted to a universal fallback trigger on the evidence of one carrier: heterogeneity across carriers of the same family, at the same bit width and under the same shift, was the largest effect we measured. And any candidate signal should be adjudicated against the trivial structural baseline — block depth — before it is called a signal at all, since here that baseline explains it away. The honest next experiments are cheap and specified: a second prompt domain to test whether the 4B depth structure reappears, and an actual measurement of the fallback gate we deliberately do not claim to have evaluated.

## ARTIFACT AVAILABILITY

Code, per-layer outcome data, and analysis scripts are released: the decisive `*_hqq4_gsm8k.jsonl` cells and `r120` re-runs, the pad-masked recomputation (`masked_from_log.json`), noise-floor nuisance runs (`nuisance_4B/1.7B.json`), and the AUROC / robustness tables (`R40_AUROC_GATE_MASKED` (canonical, masked labels) with the historical `R38_AUROC_GATE` and `R39_NOISE_RANK_SYNTHESIS`). Every number cites its `repro_command`. Models are the public Qwen3 checkpoints (Yang et al., 2025); quantization uses HQQ (Badri & Shaji, 2023). No generated or synthetic evidence supports any headline claim.

## REFERENCES

- Hicham Badri and Appu Shaji. Half-quadratic quantization of large machine learning models, November 2023. URL [https://dropbox.github.io/hqq\\_blog/](https://dropbox.github.io/hqq_blog/).
- Tim Dettmers, Mike Lewis, Younes Belkada, and Luke Zettlemoyer. LLM.int8(): 8-bit matrix multiplication for transformers at scale. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. arXiv:2208.07339.
- Elias Frantar, Saleh Ashkboos, Torsten Hoefer, and Dan Alistarh. GPTQ: Accurate post-training quantization for generative pre-trained transformers. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2210.17323.
- Sahil Kadadekar. Quality is not a safety proxy under quantization. *arXiv preprint arXiv:2606.10154*, 2026.
- Xing Li, Zeyu Xing, Yiming Li, Linping Qu, Hui-Ling Zhen, Wulong Liu, Yiwu Yao, Sinno Jialin Pan, and Mingxuan Yuan. KVTuner: Sensitivity-aware layer-wise mixed-precision KV cache quantization for efficient and nearly lossless LLM inference. In *International Conference on Machine Learning (ICML)*, 2025. arXiv:2502.04420.
- Ji Lin, Jiaming Tang, Haotian Tang, Shang Yang, Wei-Ming Chen, Wei-Chen Wang, Guangxuan Xiao, Xingyu Dang, Chuang Gan, and Song Han. AWQ: Activation-aware weight quantization for LLM compression and acceleration. In *Proceedings of Machine Learning and Systems (MLSys)*, 2024. arXiv:2306.00978.
- Davide Paglieri, Saurabh Dash, Tim Rocktäschel, and Jack Parker-Holder. Outliers and calibration sets have diminishing effect on quantization of modern LLMs. *arXiv preprint arXiv:2405.20835*, 2024.

Donghyun Son et al. NSNQuant: A double normalization approach for calibration-free low-bit vector quantization of KV cache. *arXiv preprint arXiv:2505.18231*, 2025.

An Yang et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.

Zhihang Yuan, Jiawei Liu, Jiayang Wu, Dawei Yang, Qiang Wu, Guangyu Sun, Wenyu Liu, Xing-gang Wang, and Bingzhe Wu. Benchmarking the reliability of post-training quantization: a particular focus on worst-case performance. *arXiv preprint arXiv:2303.13003*, 2023.

Pengle Zhang, Jia Wei, Jintao Zhang, Jun Zhu, and Jianfei Chen. Accurate INT8 training through dynamic block-level fallback. *arXiv preprint arXiv:2503.08040*, 2025.

## A PROVENANCE AND PRE-REGISTERED GATE LEDGER

### A.1 PROVENANCE AND RECONSTRUCTION LEDGER

Table 3 collects the reconstruction and method-sensitivity notes that in the main text would otherwise sit as dense table footnotes (cf. Table 1 and Table 2, whose footnotes here keep only a one-line pointer). All values are reproduced from the registered analysis JSONs listed in the *source* column.

### A.2 PRE-REGISTERED GATE LEDGER

Table 4 is the decision record for the two pre-registered safety flags; it replaces in-prose repeats of the bands in Limitation F and §3.1.

Table 3: Provenance and method-sensitivity ledger. All intervals are over probed blocks of one model — the unit of analysis is a layer, not an independent sample (C4).

| Quantity                                               | Qwen3-4B                                | Qwen3-1.7B                          | Source / repro                                      |
|--------------------------------------------------------|-----------------------------------------|-------------------------------------|-----------------------------------------------------|
| Probed blocks                                          | 12 of 36:<br>0,1,4,7,12,17,22,27,32–35  | 11 of 28:<br>0,1,4,7,12,17,22,24–27 | fixed probe rule<br>0,1,4,7,12,17,22,27,–4,–3,–2,–1 |
| Fisher- $z$ 95% CI on $\rho$                           | [+0.031, +0.871]                        | [−0.665, +0.525]                    | r31_paper_numbers.py                                |
| Bootstrap percentile 95% CI on $\rho$                  | [−0.085, +0.921]                        | [−0.718, +0.617]                    | r31_paper_numbers.py                                |
| Permutation $p$ (two-sided, $B=20000$ )                | 0.045                                   | 0.749                               | r31_paper_numbers.py                                |
| Leave-one-block-out range of $\rho$                    | [+0.473, +0.709]                        | [−0.273, +0.188]                    | r31_paper_numbers.py                                |
| $\Delta z = z(4B) - z(1.7B)$ lower edge (8 seeds)      | 0.794, [−0.04, +0.70]                   | —                                   | R35_DELTA_Z_CI_FIX.json                             |
| $\Delta z$ upper edge (8 seeds); seeds excl. 0         | [+0.21, +1.39]; 7 of 8                  | —                                   | r35_delta_z_ci_fix.py                               |
| Fisher SE of $\Delta z$ (projected + zh)               | 0.486 (0.317)                           | —                                   | R35_DELTA_Z_CI_FIX.json                             |
| Power on $ \Delta z =0.794$ (projected)                | 0.37 (0.71)                             | —                                   | Panel B                                             |
| Smallest $ \Delta z $ at 80% power (proj.)             | 1.36 (0.89)                             | —                                   | Panel B                                             |
| Partial- $\rho$ boot. 95% CI, turnover depth           | [−0.49, +0.66]                          | —                                   | R34_PARTIAL_CI_BOOTSTRAP.json                       |
| Partial- $\rho$ boot. 95% CI, depth turnover           | [−0.15, +0.88]                          | —                                   | r34_partial_ci_bootstrap.py                         |
| Permutation $p$ (layer-resampling)                     | 0.79 / 0.18                             | —                                   | r34_partial_ci_bootstrap.py                         |
| Depth-reweighted $\rho(\tau, \delta)$ (4B)             | 0.401                                   | —                                   | R37_DEPTH_REWEIGHT_GATE3                            |
| Analysis JSON sha16                                    | 6c26aced50960e6d                        | same                                | r31_paper_numbers.py                                |
| Canonical payload digest (max $ \Delta =0$ )           | 04ac77be066fbe87                        | 1b4143950e186756                    | sha256 dump, sort_keys                              |
| Evidence-ledger audit digest                           | e6325486081538cb                        | b13c3221d8dedabc                    | frozen cross-seat id                                |
| Registered gate AUROC (SAVED pad-polluted labels, r38) | 0.743 ( $p=0.10$ ) / 0.500 ( $p=0.54$ ) | —                                   | r38_auroc_gate.py                                   |
| Registered gate AUROC (MASKED labels, r40)             | 0.800 ( $p=0.05$ ) / 0.567 ( $p=0.40$ ) | —                                   | R40_AUROC_GATE_MASKED.json                          |
| Depth-baseline AUROC (MASKED labels)                   | 0.686 / 0.500                           | —                                   | r40_auroc_gate_masked.py                            |
| Sign flips, saved vs. masked labels                    | 4 of 12                                 | 1 of 11                             | R40_AUROC_GATE_MASKED.json                          |

*Method-sensitivity notes.* (i) Bootstrap draws are draw-sensitive at these  $n$ : a prior archived draw gives [−0.099, +0.922] on 4B vs. [−0.085, +0.921] here — both contain zero (C5). (ii) Two-sided  $p$  on 4B is convention-sensitive: 0.042 asymptotic, 0.0488 under the independent-estimator re-implementation, 0.045 here; no claim is conditioned on  $p < 0.05$ . (iii) The three  $\Delta z$  rows are the per-seed envelope ( $B=40000$ ), i.e. a seed-unstable interval — why C8 calls the sign difference observed and not established. (vii) The partial- $\rho$  CIs span zero, so the partial contrast is directionally consistent but not statistically resolved. Run config / device / trial / host identifiers are withheld for double-blind review and given in the supplementary manifest.

Table 4: Pre-registered gate ledger: band rules, measured values, and verdict (computed 0 GPU from the registered cells on the *corrected pad-masked* labels; depth baseline is the trivial block-index predictor). The AUROC half is computed under masked labels as disclosed in Limitation F; the raw pad-polluted label readout is reported in Table 3 for the label-source record, and the full three-label-set table is `R40_AUROC_GATE_MASKED.json`.

| Gate / rule                          | Band definition                                                            | 4B measured / 1.7B measured (masked labels)             | Verdict                                         |
|--------------------------------------|----------------------------------------------------------------------------|---------------------------------------------------------|-------------------------------------------------|
| Pearson $r$ PASS/KILL (raw relation) | PASS $\geq 0.3$ ; MERGE $0.15 \leq r < 0.3$ ; KILL $< 0.15$                | +0.490 / +0.021                                         | PASS / KILL (2 cells only, C12)                 |
| Registered AUROC gate                | PASS: AUROC $\geq 0.85$ and $\geq 30\%$ gain; MERGE 0.7–0.85; KILL $< 0.7$ | 0.800 (5/12; perm. $p=0.05$ ) / 0.567 (6/11; $p=0.40$ ) | MERGE / KILL; NOT passed; band fragile (F, C12) |

*Notes.* Values are the AUROC half over probed blocks, labels positive iff masked degradation  $> 0$  (`R40_AUROC_GATE_MASKED.json`). At the corrected masked labels the trivial depth-baseline AUROC is 0.686 (4B) / 0.500 (1.7B) — i.e. the depth baseline does *not* beat the turnover AUROC on either carrier, correcting Limitation F. The band verdict is fragile: a single label flip moves the 4B AUROC to anywhere in  $[0.639, 0.889]$ , spanning KILL/MERGE/PASS, and the 4B mask label count rests on one near-zero layer (C12). The gate’s damage-gain half ( $\geq 30\%$ ) needs a fallback-baseline simulation and was not run (C7). Both rules were defined over *eight* cells of which only two were run (C12), so both classifications are boundary observations, not passes or kills of the rules.